TRACK: UX/UI DESIGN · WEEK 6 · BUILD+

UX-06D: The Edge Case & Empty State

10 Nightmare Scenarios · 6 Mandatory States · Plain-English Diagnostics · 60s Undo Rollback

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Project: FlyCommerce OS

NIGHTMARE SCENARIOS	MANDATORY UI STATES	ERROR COPY STANDARD	LOADING FIDELITY
10 Operational Cases	6 Production Designs	Plain English (No 500s)	True 40px Shimmer Skeleton

1. Six Mandatory Production UI States

1. Action-Oriented Empty State

"*No Active SKUs Matching 'SKU-UNKNOWN'.*"
Provides 2 concrete next steps:
• `[ Clear All Filters ]`
• `[ Add New Product SKU ]`

2. True Structural Skeleton

Exact 1:1 geometry of 40px high-density table rows with pulsing gradient shimmer blocks; zero layout shift on data load.

3. Plain-English Error Toast

Replaces "Error 504" with: "*Unable to Lock TikTok Buffer. Wholesale inventory safe and unchanged.*" + `[ Retry (Attempt 2) ]`.

4. Success with 60s Undo

Floating dark toast with animated SVG countdown: "*Rebalanced 150u to TikTok Shop*" + `[ Undo Allocation (58s) ]`.

5. Branded 404 Recovery

"*This Inventory Route Does Not Exist.*"
Embedded search bar + `[ Return to Live Stock Grid ]` CTA.

6. Overflow & Truncation

220px text clamp with hover tooltip + 99+ alert aggregation grouping 14 variants into an Executive Threat Card.

2. The 10 "Nightmare" Operational Edge Cases

Scenario	Trigger Event	Operational Risk	FlyCore UI Mitigation
N-01: API Blackout	TikTok Shop webhook gateway HTTP 503 during sales.	Overselling stock; seller health score demotion.	Global persistent amber warning; 25 safety buffer units automatically locked.
N-02: Zero Search Results	User misspells SKU (`GLW-SERUM-01`).	Operator assumes SKU deleted; wasted time.	Fuzzy-match suggestion ("Did you mean...") with 1-click filter reset.
N-03: Alert Flooding	Viral surge triggers 14 variant alerts at once.	Cognitive overload and alert blindness.	Notification clustering: Bundles 14 items into an Executive Threat Card.
N-04: 500-Char Title	Supplier uploads keyword-stuffed title with emoji.	Table row explodes from 40px to 300px.	CSS clamp at 280px with hover tooltip and 1-click clipboard copy.
N-05: Reserve Depletion	Wholesale pool reaches 0 units.	Rebalance modal dead-end; stockout penalty.	AI suggests alternative reservoir: Amazon FBA split or expediting PO.
N-06: Corrupted CSV	Batch upload has trailing spaces in SKU names.	Crashes warehouse WMS database pipeline.	Pre-flight client-side linter flags rows with 1-click "Auto-Sanitize".
N-07: Wi-Fi Dead Zone	Picker tablet enters warehouse cold-storage.	Scanned orders lost in memory.	Offline Mode Banner: 14 scans saved in IndexedDB; auto-sync on Wi-Fi.
N-08: Concurrency Lock	Elena & Marcus edit same SKU within 500ms.	Silent state overwrite.	Non-destructive merge alert: "Marcus updated stock 3s ago; merge or keep?".
N-09: Broken 404 Link	User clicks expired report URL from Slack.	User hits raw browser error screen.	Branded 404 recovery screen with direct shortcuts to Live Stock Grid.
N-10: Numeric Overflow	Stock quantity reaches 14,250,000 units.	Text overflows, pushing buttons off-screen.	Tabular numeric formatting (14.25M Units) expanding on hover.

Master Figma Workspace: https://www.figma.com/file/r7K8x9pL4mN2qS/FlyRank-UX-UI-Design-Capstone?type=design&node-id=0%3A1&mode=design&t=review-access
GitHub Repository: https://github.com/KMSiam/Flyrank-Internship-UX-UI-Design